

N-CASA AUTOMATED SECURITY AUDIT REPORT

Network Configuration Automated Security Auditor — Executive Audit Snapshot

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Compliance Framework: CIS

Audit Status: AI_ANALYSIS_COMPLETE

Report Version: 1.0

MANDATORY SAFETY NOTICE & AUTHORIZATION REQUIREMENT

All remediation commands contained within this report are proposals only. They were NOT automatically executed on network devices. Manual validation, peer review, and authorization are required prior to applying any configuration changes to production environments.

AI ADVISORY NOTICE

AI-assisted analysis included in this report is advisory. Deterministic compliance results and security rules remain authoritative.

1. EXECUTIVE SUMMARY

METRIC	VALUE	METRIC	VALUE
Files Audited	6	Controls Evaluated	52
Controls Passed (PASS)	20	Open Findings (FAIL)	2
Scope Limitations (NOT_VERIFIABLE)	30	Remediation Proposals	1

2. COMPLIANCE EVALUATION SUMMARY

Framework Breakdown

FRAMEWORK	PASS	FAIL	NOT_VERIFIABLE
CIS	12	0	20
NIST	7	2	7
STIG	1	0	3

Detailed Compliance Control Results

CONTROL ID	FRAMEWORK	CONTROL TITLE	SEVERITY	STATUS
CIS-NET-SSH-001	CIS	Secure SSH Management Configuration	HIGH	PASS
CIS-NET-TELNET-001	CIS	Disable Insecure Telnet Management	HIGH	NOT_VERIFIABLE
CIS-NET-HTTP-001	CIS	Disable Unencrypted HTTP Management Server	HIGH	NOT_VERIFIABLE
CIS-NET-HTTPS-001	CIS	Enable Encrypted HTTPS Web Management	MEDIUM	PASS
CIS-NET-ID-001	CIS	Device Hostname and Identification	LOW	PASS
CIS-NET-IFACE-001	CIS	Administrative Shutdown of Unused Interfaces	MEDIUM	NOT_VERIFIABLE
CIS-NET-FW-001	CIS	Firewall Policy Action Explicitness	HIGH	PASS

CONTROL ID	FRAMEWORK	CONTROL TITLE	SEVERITY	STATUS
CIS-NET-ACL-001	CIS	ACL Rule Action Explicitness	MEDIUM	NOT_VERIFIABLE
NIST-NET-AUTH-001	NIST	Centralized Authentication and AAA Framework	HIGH	NOT_VERIFIABLE
NIST-NET-LOG-001	NIST	Centralized Syslog Event Audit Logging	MEDIUM	PASS
NIST-NET-NTP-001	NIST	Network Time Protocol Synchronization	MEDIUM	PASS
NIST-NET-VPN-001	NIST	Virtual Private Network Configuration Visibility	MEDIUM	NOT_VERIFIABLE
STIG-NET-ROUTING-001	STIG	Dynamic Routing Protocol Configuration Visibility	MEDIUM	NOT_VERIFIABLE
CIS-NET-SSH-001	CIS	Secure SSH Management Configuration	HIGH	NOT_VERIFIABLE
CIS-NET-TELNET-001	CIS	Disable Insecure Telnet Management	HIGH	NOT_VERIFIABLE
CIS-NET-HTTP-001	CIS	Disable Unencrypted HTTP Management Server	HIGH	NOT_VERIFIABLE
CIS-NET-HTTPS-001	CIS	Enable Encrypted HTTPS Web Management	MEDIUM	NOT_VERIFIABLE
CIS-NET-ID-001	CIS	Device Hostname and Identification	LOW	PASS
CIS-NET-IFACE-001	CIS	Administrative Shutdown of Unused Interfaces	MEDIUM	NOT_VERIFIABLE

CONTROL ID	FRAMEWORK	CONTROL TITLE	SEVERITY	STATUS
CIS-NET-FW-001	CIS	Firewall Policy Action Explicitness	HIGH	NOT_VERIFIABLE
CIS-NET-ACL-001	CIS	ACL Rule Action Explicitness	MEDIUM	NOT_VERIFIABLE
NIST-NET-AUTH-001	NIST	Centralized Authentication and AAA Framework	HIGH	FAIL
NIST-NET-LOG-001	NIST	Centralized Syslog Event Audit Logging	MEDIUM	FAIL
NIST-NET-NTP-001	NIST	Network Time Protocol Synchronization	MEDIUM	NOT_VERIFIABLE
NIST-NET-VPN-001	NIST	Virtual Private Network Configuration Visibility	MEDIUM	NOT_VERIFIABLE
STIG-NET-ROUTING-001	STIG	Dynamic Routing Protocol Configuration Visibility	MEDIUM	NOT_VERIFIABLE
CIS-NET-SSH-001	CIS	Secure SSH Management Configuration	HIGH	PASS
CIS-NET-TELNET-001	CIS	Disable Insecure Telnet Management	HIGH	NOT_VERIFIABLE
CIS-NET-HTTP-001	CIS	Disable Unencrypted HTTP Management Server	HIGH	NOT_VERIFIABLE
CIS-NET-HTTPS-001	CIS	Enable Encrypted HTTPS Web Management	MEDIUM	NOT_VERIFIABLE
CIS-NET-ID-001	CIS	Device Hostname and Identification	LOW	PASS

CONTROL ID	FRAMEWORK	CONTROL TITLE	SEVERITY	STATUS
CIS-NET-IFACE-001	CIS	Administrative Shutdown of Unused Interfaces	MEDIUM	NOT_VERIFIABLE
CIS-NET-FW-001	CIS	Firewall Policy Action Explicitness	HIGH	PASS
CIS-NET-ACL-001	CIS	ACL Rule Action Explicitness	MEDIUM	NOT_VERIFIABLE
NIST-NET-AUTH-001	NIST	Centralized Authentication and AAA Framework	HIGH	NOT_VERIFIABLE
NIST-NET-LOG-001	NIST	Centralized Syslog Event Audit Logging	MEDIUM	PASS
NIST-NET-NTP-001	NIST	Network Time Protocol Synchronization	MEDIUM	PASS
NIST-NET-VPN-001	NIST	Virtual Private Network Configuration Visibility	MEDIUM	NOT_VERIFIABLE
STIG-NET-ROUTING-001	STIG	Dynamic Routing Protocol Configuration Visibility	MEDIUM	NOT_VERIFIABLE
CIS-NET-SSH-001	CIS	Secure SSH Management Configuration	HIGH	PASS
CIS-NET-TELNET-001	CIS	Disable Insecure Telnet Management	HIGH	PASS
CIS-NET-HTTP-001	CIS	Disable Unencrypted HTTP Management Server	HIGH	NOT_VERIFIABLE
CIS-NET-HTTPS-001	CIS	Enable Encrypted HTTPS Web Management	MEDIUM	NOT_VERIFIABLE

CONTROL ID	FRAMEWORK	CONTROL TITLE	SEVERITY	STATUS
CIS-NET-ID-001	CIS	Device Hostname and Identification	LOW	PASS
CIS-NET-IFACE-001	CIS	Administrative Shutdown of Unused Interfaces	MEDIUM	PASS
CIS-NET-FW-001	CIS	Firewall Policy Action Explicitness	HIGH	NOT_VERIFIABLE
CIS-NET-ACL-001	CIS	ACL Rule Action Explicitness	MEDIUM	NOT_VERIFIABLE
NIST-NET-AUTH-001	NIST	Centralized Authentication and AAA Framework	HIGH	PASS
NIST-NET-LOG-001	NIST	Centralized Syslog Event Audit Logging	MEDIUM	PASS
NIST-NET-NTP-001	NIST	Network Time Protocol Synchronization	MEDIUM	PASS
NIST-NET-VPN-001	NIST	Virtual Private Network Configuration Visibility	MEDIUM	NOT_VERIFIABLE
STIG-NET-ROUTING-001	STIG	Dynamic Routing Protocol Configuration Visibility	MEDIUM	PASS

3. ACTIONABLE SECURITY FINDINGS

FND-AUD-88ACE7-NIST-NET-AUTH-001: Centralized Authentication and AAA Framework	HIGH	OPEN
Control ID: NIST-NET-AUTH-001 (NIST)		
Description: Verify that AAA authentication framework and privileged secrets are configured.		

Expected: Centralized authentication (AAA) or secure encrypted administrative secrets must be configured. | **Observed:** AAA framework is explicitly disabled and no encrypted enable secret <REDACTED>

FND-AUD-88ACE7-NIST-NET-LOG-001: Centralized Syslog Event Audit Logging MEDIUM OPEN

Control ID: NIST-NET-LOG-001 (NIST)

Description: Verify that remote syslog servers are configured for audit event logging.

Expected: Device audit event logging must be forwarded to centralized remote Syslog servers. | **Observed:** Audit event logging is explicitly disabled

4. ASSESSMENT LIMITATIONS (NOT_VERIFIABLE)

CONTROL ID	FRAMEWORK	TITLE	REASON / REQUIRED VERIFICATION
CIS-NET-TELNET-001	CIS	Disable Insecure Telnet Management	Telnet management state is not specified
CIS-NET-HTTP-001	CIS	Disable Unencrypted HTTP Management Server	HTTP management server state is not specified
CIS-NET-IFACE-001	CIS	Administrative Shutdown of Unused Interfaces	All 2 interface(s) are active/enabled or unverified
CIS-NET-ACL-001	CIS	ACL Rule Action Explicitness	No Access Control Lists (ACLs) configured.
NIST-NET-AUTH-001	NIST	Centralized Authentication and AAA Framework	AAA configuration state is unverified
NIST-NET-VPN-001	NIST	Virtual Private Network Configuration Visibility	No structured VPN configuration detected
STIG-NET-ROUTING-001	STIG	Dynamic Routing Protocol Configuration Visibility	No dynamic routing or static route entries detected
CIS-NET-SSH-001	CIS	Secure SSH Management Configuration	No SSH configuration status present

CONTROL ID	FRAMEWORK	TITLE	REASON / REQUIRED VERIFICATION
CIS-NET-HTTPS-001	CIS	Enable Encrypted HTTPS Web Management	HTTPS web management server state is not specified / disabled
CIS-NET-FW-001	CIS	Firewall Policy Action Explicitness	No firewall security policies configured on device.
NIST-NET-NTP-001	NIST	Network Time Protocol Synchronization	No NTP servers configured in parsed scope

5. PROPOSED REMEDIATION STEPS

REM-AUD-88ACE7-FND-AUD-88ACE7-NIST-NET-AUTH-001: Enable AAA Authentication Architecture

Vendor: **Cisco** | Device Type: **Router / Switch** | Status: **AVAILABLE**

Proposed Configuration Commands:

```
aaa new-model
```

Validation Steps:

- Review the proposed configuration snippet with an authorized network engineer.
- Apply the configuration change in a non-production test environment.
- Re-run N-CASA compliance evaluation for the target device configuration.
- Verify that control NIST-NET-AUTH-001 status changes from FAIL to PASS.

REM-AUD-88ACE7-FND-AUD-88ACE7-NIST-NET-LOG-001: Manual Review Required for Centralized Syslog Event Audit Logging

Vendor: **Cisco** | Device Type: **Network Device** | Status: **MANUAL_REVIEW_REQUIRED**

MANUAL REVIEW REQUIRED: Vendor-specific automatic CLI generation is not available for unsupported or unknown vendors.

Validation Steps:

- Review the proposed configuration snippet with an authorized network engineer.
- Apply the configuration change in a non-production test environment.
- Re-run N-CASA compliance evaluation for the target device configuration.
- Verify that control NIST-NET-LOG-001 status changes from FAIL to PASS.

6. AI-ASSISTED CONFIGURATION ANALYSIS

The following analysis was generated by AI for unknown configuration syntax. Deterministic rules remain authoritative.

Metric	Value
Analysis Status	AI_ANALYSIS_COMPLETE
Analyzed Files	6

7. DEVICE & CONFIGURATION INVENTORY

File Name	Vendor	Device Type	Confidence	Method	Line Count
ncasa_manual_test_configs/README.txt	Unknown	Unknown	100%	DETERMINISTIC	N/A
ncasa_manual_test_configs/02_fortinet_firewall_ruleset.fir	Fortinet	rt: Firewall	79%	DETERMINISTIC	N/A
ncasa_manual_test_configs/02_cisco_routers/router_insert_line.cfg	Cisco	Unknown router	71%	DETERMINISTIC	N/A
ncasa_manual_test_configs/03_juniper_firewall_ruleset.fir	Juniper	ni: Firewall	92%	DETERMINISTIC	N/A
ncasa_manual_test_configs/01_cisco_routers/router_security.cfg	Cisco	Router	85%	DETERMINISTIC	N/A
ncasa_manual_test_configs/04_unknown_device_config.cfg	Unknown	kn: Unknown	100%	DETERMINISTIC	N/A